

Label all graphs with the 'key features' (x- and y-intercepts, vertical and horizontal asymptotes)

1. Sketch the graphs of $f(x)$, $g(x)$, $h(x)$ and answer the appropriate question for each function.

a) $f(x) = \frac{-2}{x^2 - 4x + 4} = \frac{-2}{(x-2)^2}$

(4)

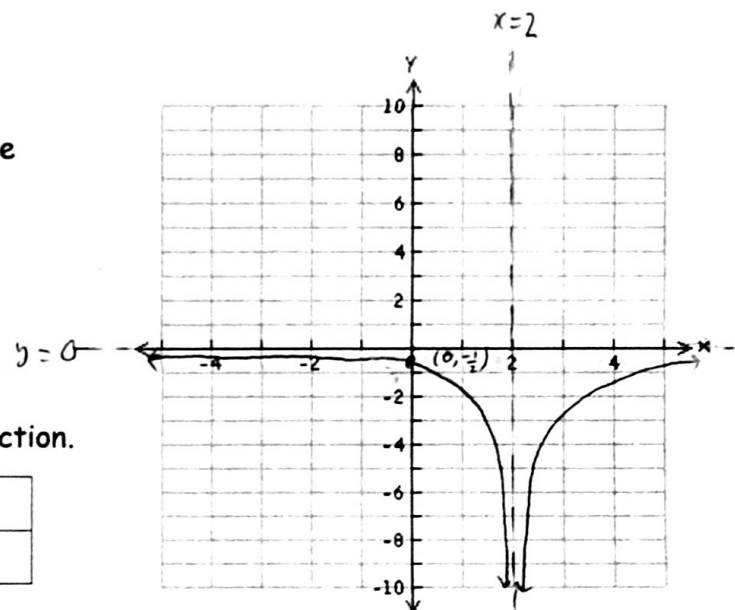
- i) determine how the graph approaches the vertical asymptote(s).

as $x \rightarrow 2^-$, $f(x) \rightarrow -\infty$

as $x \rightarrow 2^+$, $f(x) \rightarrow -\infty$

- ii) determine the end behavior for the function.

as $x \rightarrow -\infty$	$f(x) \rightarrow 0^-$
as $x \rightarrow \infty$	$f(x) \rightarrow 0^-$

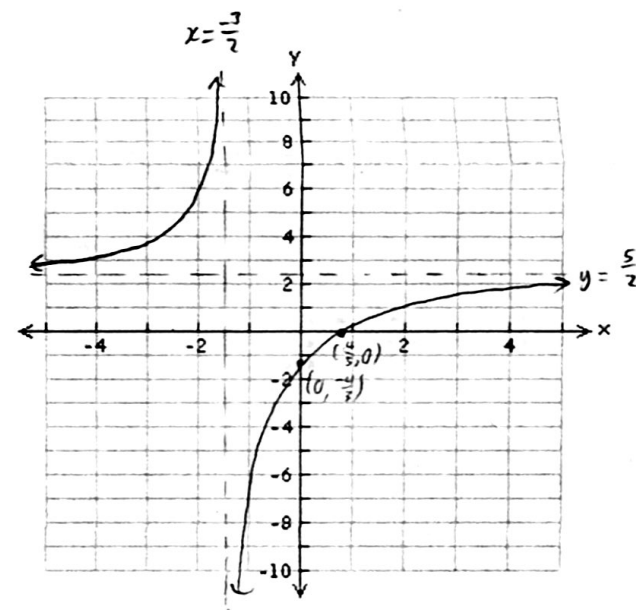


b) $g(x) = \frac{5x-4}{3+2x}$

(4)

- iii) Complete the table.

Interval	$(-\infty, -\frac{3}{2})$	$(-\frac{3}{2}, \frac{4}{5})$	$(\frac{4}{5}, \infty)$
Sign of Function	+	-	+
Sign of Slope (+/-)	+	+	+
Change in Slope (increasing/decreasing)	↑	↓	↓



c) $h(x) = \frac{2}{x^2 + x - 12} = \frac{2}{(x+4)(x-3)}$

(4)

- iv) Complete the table, no decimals!

vertical asymptote(s)	$x = -4, x = 3$
horizontal asymptote(s)	$y = 0$
x-intercept(s)	None
y-intercept	$(0, -\frac{1}{6})$
range	$\{y \in \mathbb{R} \mid y \leq -\frac{8}{49}, y > 0\}$

$\frac{3+(-4)}{2} = -\frac{1}{2}$ $h(-\frac{1}{2}) = \frac{2}{(-\frac{1}{2}+4)(-\frac{1}{2}-3)}$

$= -\frac{8}{49}$

